

WASP-35b

R-1148

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_241202 · created 2026-07-31T00:52:13+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460646.895 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.017
Transit depth (Rp/Rs)^2	1.91 +/- 0.48 [%]
Orbital Inclination (inc)	89.61 +/- 1.2
Airmass coefficient 1 (a1)	1.0285 +/- 0.0078
Airmass coefficient 2 (a2)	-0.0123 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	#3 - [433, 237]
Optimal Aperture	4.74
Optimal Annulus	12.87
Transit Duration (day)	0.0586 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.138 $\pm$ 0.017 vs 0.1241 (deviation 0.82 σ)
Duration fit vs published	0.0586 d vs 0.1317 d (deviation 17.82 σ)
Mid-time O-C	-4.1 min
Depth SNR	8.1
Residual scatter	0.76 %

Light curve

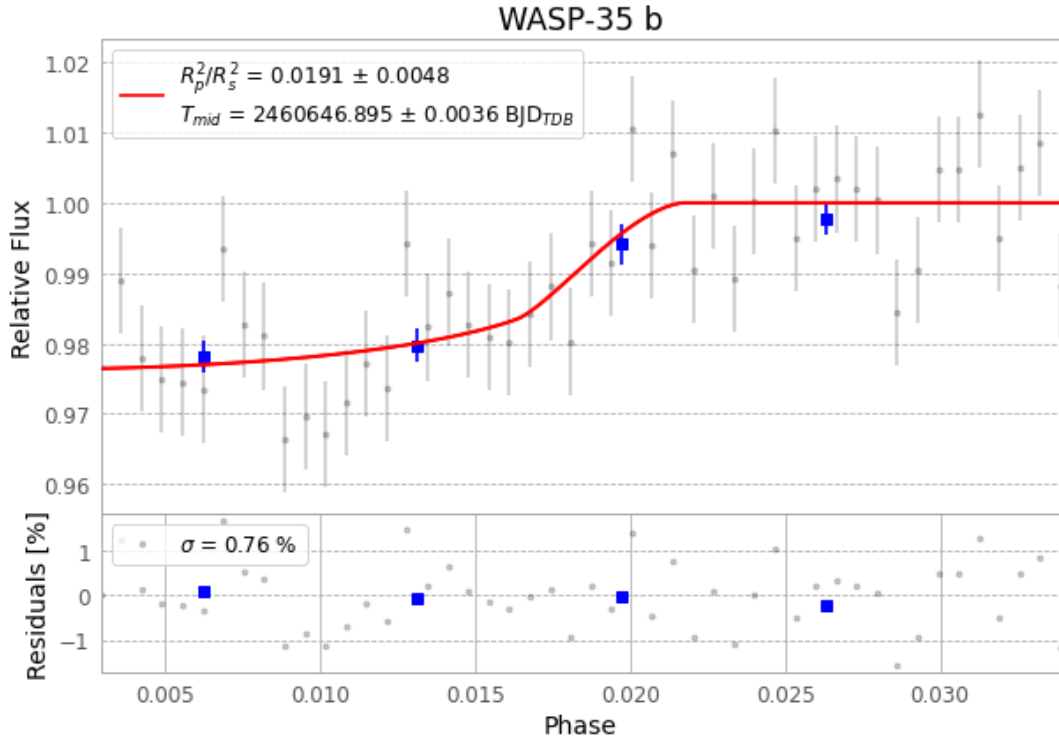


Figure 1 — FinalLightCurve_WASP-35 b_2024-12-02.png (SHA-256 14ae10dab76b937e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0276381dedaa2bf7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

48 FITS frames (32 MB), WASP-35241202093317.FITS → WASP-35241202115416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-241202.log (363b05ffb73e...), FinalLightCurve_WASP-35 b_2024-12-02.png (14ae10dab76b...), FinalLightCurve_WASP-35 b_2024-12-02.csv (48d54bfaa8e8...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 00:52Z.